Simplifying the First-Stage MILP: Reduced Formulation, Justification, and COPT Verification

Manuscript: Drone Failure-Aware Truck-Drone Collaborative Routing for Multi-Type Coverage Tasks (second revision). Equation numbers refer to the R1 PDF: objective (1) and constraints (2)-(43) of the first-stage model; (47)-(49) are the linearisation constraints of Section 4.3.

1. Summary

Result. The 42 first-stage constraints (2)-(43) can be reduced to **28 numbered constraints** without changing the set of executable plans or the optimal value:

- **9 constraints are removed**: (7), (8), (10), (11), (14), (17), (21), (22), (26);
- **(9) is replaced** by an unconditional bound (9'), which also removes the auxiliary variable δ_s^d and its linearisation (47)-(49);
- **(12)-(13) are extended** to point targets, which makes (21)-(22) redundant;
- **5 pairs are written as two-sided constraints**: (5)+(6), (31)+(32), (33)+(34), (38)+(39), (40)+(41).

Three corrections are needed independently of the reduction: (2) becomes $\sum_{i \in N_2} x_{0i} = 1$, and the domains of (18) and (30) are extended to the depots. Without them the printed model accepts plans that the truck cannot execute.

Evidence. (i) All 105 stored SVNS plans satisfy every constraint. (ii) On two test instances solved to proven optimality, the reduced and full models have identical optima, whereas removing (16), (23) or (37) produces unexecutable plans. (iii) With 30-minute COPT runs, both models reach the Gurobi values of R1 on C-4-3 (7.4799) and R-4-3 (7.4748). On RC-4-3 both reach 7.9140, 0.24% above the Gurobi value 7.895, with lower bounds below 7.895. Every returned plan is executable, and the full model accepts the reduced model's plans. Solve times vary between runs; overall the two models are comparable in speed.

2. Corrections to the printed model (required before reduction)

The model as printed in R1 contains three domain errors. The original Gurobi implementation already contained the corrected forms, so the reported Gurobi values are not affected, but the text of the paper must be corrected. All results below use the corrected model as the reference.

2.1 Depot constraint (2)

$$\sum_{i \in N_2} x_{0i} = \sum_{i \in N_1} x_{i|N|} \qquad \text{(2) printed}$$

$$\sum_{i \in N_2} x_{0i} = 1 \qquad \text{(2') corrected}$$

Reason. The printed constraint only equates outflow from the origin with inflow to the destination. It is satisfied both by a truck that never leaves (all $x_{ij} = 0$) and by several departures. Constraint (2') fixes exactly one departure. Summing flow conservation (3) over N_p then gives exactly one arrival at the destination, so the printed (2) is implied and can be dropped.

2.2 Node-sortie link (18): domain extended to the depots

$$z^{sd}_{jab} + z^{sd}_{iab} \geq y^{sd}_{ai} + y^{sd}_{jb} - M\left(2 - y^{sd}_{ai} - y^{sd}_{jb}\right) \qquad (18')$$
$$\forall\, s \in S, d \in D, a \in N_1, b \in N_2, i, j \in N_s \left(printed: a, b \in N_p\right)$$

Reason. The printed domain $a, b \in N_p$ omits every sortie that starts at the origin or ends at the destination. For such sorties z^{sd}_{iab} can be assigned to a wrong launch-retrieval pair, and the coordination constraint (23), which requires the truck to visit a and b, is bypassed. In the 30 SVNS plans, 186 of 401 sorties start at the origin or end at the destination. On instance TOY_1D the printed model returns 7.378 instead of the true optimum 7.908, with a drone retrieved at a node that the truck never visits.

2.3 Drone launch time (30): domain extended to the origin

$$\hat{t}^d_l \geq \left(1 - R^d_l\right)\left(\check{t}_l + LTD\right) + R^d_l\left(\check{t}^d_l + LTD\right) - M\left(1 - L^d_l\right), \forall\, l \in N_1, d \in D \qquad (30')$$
$$L^d_l = \sum_{s \in S}\sum_{j \in N_s} y^{sd}_{lj},\ R^d_l = \sum_{s \in S}\sum_{i \in N_s} y^{sd}_{il},\ R^d_0 = 0$$

Reason. The printed domain is $l \in N_p$, so a drone launched at the origin has no launch-time constraint at all. The 30 SVNS plans contain 80 such launches. The products $R^d_l \check{t}_l$ and $R^d_l \check{t}^d_l$ are linearised exactly, as two big-M rows for the cases $R^d_l = 0$ and $R^d_l = 1$.

3. Proposed simplifications

Table 1 lists every change. In the reason column, 'implied' means that every solution of the reduced model satisfies the removed constraint, or can be turned into one that does without changing the routes or the objective. Sections 3.1-3.7 give the formulas and the arguments.

Table 1. Overview of the reduction (equation numbers of the R1 PDF).

Original	Action	Replacement	Reason (one line)
(7)	remove	-	implied by (5)-(6) for every precedence variable used in (37)
(8), (10), (11)	remove	-	implied by flow conservation (13), MTZ (15) and (9')
(9)	replace	(9'): $\sum_{a \in N_1}\sum_{j \in N_s} y^{sd}_{aj} \leq 1$	same meaning once (16) holds; removes δ^d_s and (47)-(49)
(14)	remove	domain $w^{sd}_i \in \mathbb{Z}_+$	MTZ (15) needs no bounds on w
(17)	remove	-	$\sum_d (binary) \leq \lvert D \rvert \leq \bar{v}$ always
(21), (22)	remove	(12'), (13') with $s \in S$	(21)-(22) are (12)-(13) for $\lvert N_s \rvert = 1$
(26)	remove	-	waiting before arrival can be moved to the previous departure
(5)+(6), (31)+(32), (33)+(34), (38)+(39), (40)+(41)	merge	two-sided rows	identical feasible set, fewer equation numbers
(2), (18), (30)	correct	(2'), (18'), (30')	domain errors, see Section 2

3.1 Remove (7): precedence symmetry is implied by (5)-(6)

$$u_i - u_j \geq 1 - (C+2)\, p_{ij}, \forall\, i \in N_1, j \in N_2, i \neq j \quad (5)$$

$$u_i - u_j \leq -1 + (C+2)(1 - p_{ij}), \forall\, i \in N_1, j \in N_2, i \neq j \quad (6)$$

$$p_{ij} + p_{ji} = 1, \forall\, i, j \in N_3, i \neq j \quad (7) \text{ removed}$$

Reason. For binary p_{ij}, (5)-(6) state that $p_{ij}=1 \Rightarrow u_i < u_j$ and $p_{ij}=0 \Rightarrow u_i > u_j$. If both (i, j) and (j, i) belong to $N_1 \times N_2$, then $p_{ij}=1$ forces $u_i < u_j$, which rules out $p_{ji}=1$; likewise $p_{ij}=0$ forces $p_{ji}=1$. Hence $p_{ij} + p_{ji} = 1$ already holds. The only precedence variables not fixed by (5)-(6) are p_{a0}. They appear only in rows of (37) whose second sortie is launched at the origin. With (7) these rows are switched off ($p_{a0}=1 - p_{0a}=0$) in every executable plan, because the origin precedes every visited node. Without (7) the solver can also switch them off, so nothing changes. Removing (7) deletes $|N_3|(|N_3|-1)$ equality rows (56 rows on C-4-3).

3.2 Replace (8)-(11) by one bound (9'); drop δ and (47)-(49)

$$1 - M(1 - \delta_s^d) \leq \sum_{i \in N_1} \sum_{j \in N_s} y_{ij}^{sd} \leq 1 + M(1 - \delta_s^d) \quad (8),(9)$$

$$1 - M(1 - \delta_s^d) \leq \sum_{i \in N_s} \sum_{j \in N_2} y_{ij}^{sd} \leq 1 + M(1 - \delta_s^d) \quad (10),(11)$$

$$\sum_{a \in N_1} \sum_{j \in N_s} y_{aj}^{sd} \leq 1, \forall\, s \in S \setminus S_{node}, d \in D \quad (9') \text{ new}$$

Reason. Fix $s \in S \setminus S_{node}$ and $d \in D$. Let A^{sd} be the number of launch arcs ($N_1 \to N_s$) and B^{sd} the number of retrieval arcs ($N_s \to N_2$) of drone d for target s.

- $A^{sd} = B^{sd}$ **always.** Summing flow conservation (13) over all $p \in N_s$ cancels every internal arc. Hence (10) ⇔ (8) and (11) ⇔ (9).
- **(8) is implied.** If drone d uses an internal arc of target s ($\delta_s^d = 1$), the internal arcs form paths because (15) excludes cycles. The first node of such a path is entered from N_1, so $A^{sd} \geq 1$.
- **(9) becomes unconditional.** Constraint (16) requires every sweep pair $(i, i+1)$ of an area or line target to be flown consecutively, and every coverage node of such a target belongs to a sweep pair. So any drone that serves s uses an internal arc, i.e. $\delta_s^d = 1$, and the gated bound (9) is equivalent to (9'). When d does not serve s, (9') holds trivially because $A^{sd} = 0$.

With (9') no constraint uses δ_s^d any more, so these variables and the linearisation (47)-(49) are removed as well. (9') still guarantees that each drone serves each target in at most one sortie, as stated in R1.

3.3 Remove (14): the MTZ constraint (15) does not need bounds

$$1 \leq w_i^{sd} \leq |N_s| \quad (14) \text{ removed}$$

$$w_i^{sd} - w_j^{sd} + |N_s|\, y_{ij}^{sd} \leq |N_s| - 1 \quad (15) \text{ kept}$$

Reason. Subtour elimination comes from (15) alone: along any cycle of used arcs, (15) gives $w_j^{sd} \geq w_i^{sd} + 1$ for every arc, which is impossible. Conversely, for any feasible sortie, setting w_i^{sd} equal to the visit position $1, 2, \ldots, |N_s|$ satisfies both (14) and (15), so (14) never removes a valid route. It is enough to declare $w_i^{sd} \in \mathbb{Z}_+$ in the variable definitions. On the C-4-3 solution of the reduced model, (14)

is violated in some rows, but only because w lies outside $[1, |N_s|]$. The routes and the objective are unchanged.

3.4 Remove (17): capacity can never bind

$$\sum_{d \in D} \max_{i,j} \{ y_{ij}^{sd} \} \le \bar{v}, \forall s \in S \qquad \text{(17) removed}$$

Reason. The left-hand side is a sum of $|D|$ binary terms, so it never exceeds $|D|$. D is the set of drones carried by the truck, so $|D| \le \bar{v}$ (in all instances $|D| = \bar{v}$). The constraint therefore never binds, and its max-linearisation rows are removed as well. If a future variant allowed $|D| > \bar{v}$, (17) would have to be kept.

3.5 Remove (21)-(22) by extending (12)-(13) to point targets

$$\sum_{d \in D} \sum_{i \in N_s \cup N_1, i \ne j} y_{ij}^{sd} = 1, \forall s \in S, j \in N_s \qquad (12')$$

$$\sum_{i \in N_s \cup N_1, i \ne p} y_{ip}^{sd} = \sum_{j \in N_s \cup N_2, j \ne p} y_{pj}^{sd}, \forall s \in S, d \in D, p \in N_s \qquad (13')$$

Reason. For a point target, $N_s = \{j\}$ contains one node, so $(N_s \cup N_1) \setminus \{j\} = N_1$ and $(N_s \cup N_2) \setminus \{j\} = N_2$. Then (12') and (13') are literally (21) and (22). Changing the domain from $s \in S \setminus S_{node}$ to $s \in S$ therefore removes two constraints without changing the model. **Caution:** deleting (21)-(22) without extending (12)-(13) is wrong. On TOY_1D it gives a 5.5% lower, unexecutable plan (Figure 3).

3.6 Remove (26): truck waiting can be moved to the previous departure

$$\check{t}_j \ge \hat{t}_i + t_{ij} - M(1 - x_{ij}), \forall i \in N_1, j \in N_2 \qquad \text{(25) kept}$$

$$\check{t}_j \le \hat{t}_i + t_{ij} + M(1 - x_{ij}), \forall i \in N_1, j \in N_2 \qquad \text{(26) removed}$$

Reason. Constraint (25) already says that the truck cannot reach node j before it leaves node i and travels to j. Constraint (26) additionally forbids arriving later. A later arrival only means that the truck waits on the way, and the same waiting can be spent at node i before departure. The routes and the completion time stay the same, so removing (26) never changes the optimal value.

3.7 Merge one-sided pairs into two-sided constraints

$$1 - (C+2) p_{ij} \le u_i - u_j \le C + 1 - (C+2) p_{ij} \qquad \text{(5') = (5)+(6)}$$

$$|\hat{t}_j^d - \check{t}_j^d - t_j^s| \le M\left(1 - \sum_{i \in N_s \cup N_1} y_{ij}^{sd}\right) \qquad \text{(31') = (31)+(32)}$$

$$|\check{t}_j^d - \hat{t}_i^d - t_{ij}^d| \le M(1 - y_{ij}^{sd}) \qquad \text{(33') = (33)+(34)}$$

$$T K_j^{min} - M\left(1 - \sum_{i \in N_1} x_{ij}\right) \le \check{t}_j \le T K_j^{max} + M\left(1 - \sum_{i \in N_1} x_{ij}\right) \qquad \text{(38') = (38)+(39)}$$

$$T D_s^{min} - M\left(1 - \sum_{i} y_{ij}^{sd}\right) \le \check{t}_j^d \le T D_s^{max} + M\left(1 - \sum_{i} y_{ij}^{sd}\right) \qquad \text{(40') = (40)+(41)}$$

These are notational merges: the domains are unchanged and the feasible set is identical. They cut five equation numbers.

3.8 Constraints that must be kept

- **(16), (23), (37)** look like candidates, because (37) is never tight in the SVNS plans. However, removing any of them lets COPT return a lower objective with an unexecutable plan (Figure 3). Without (16) a sweep line is not flown consecutively, without (23) launches and retrievals happen at nodes the truck does not visit, and without (37) one drone flies two sorties at the same time.
- **(38)-(43)** never bind on the benchmark, because the time windows are $[0, 50]$, $D_D = 15$ and $T_D = 200$. They define the problem class, though, and must be kept. We recommend stating in the paper that they are not binding on the reported instances.

3.9 Resulting reduced model

Table 2. Constraint list of the reduced first-stage model (primes mark corrected, replaced or merged constraints).

Block	Constraints in the reduced model	Count
Depot	(2')	1
Truck routing and precedence	(3), (4), (5')	3
Drone routing	(9'), (12'), (13'), (15), (16), (18'), (19), (20)	8
Truck-drone coordination	(23), (24)	2
Truck timing	(25), (27), (28), (29)	4
Drone timing	(30'), (31'), (33'), (35), (36), (37)	6
Time windows	(38'), (40')	2
Route and sortie duration	(42), (43)	2
Total		**28** (R1: 42)

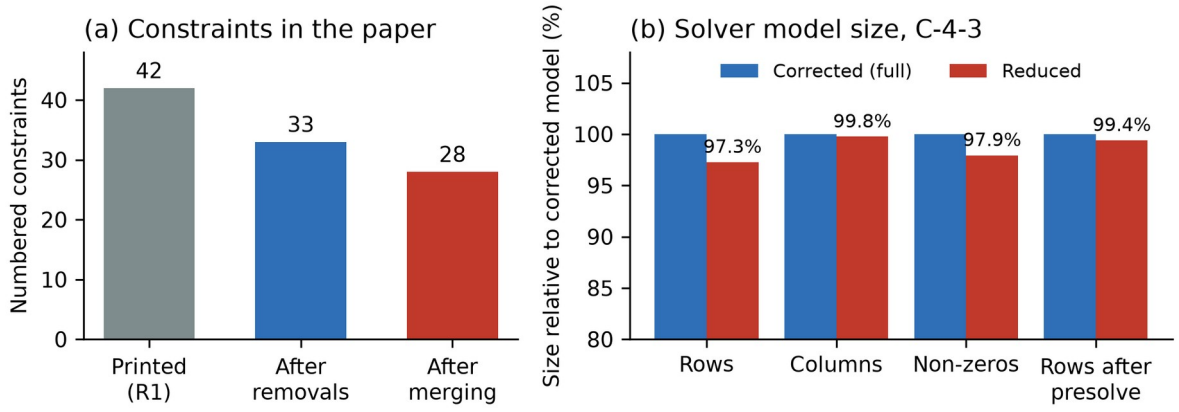


Figure 1. (a) Number of numbered first-stage constraints. (b) Size of the solver model for C-4-3 after linearisation. The reduction mainly simplifies the paper; the solver model shrinks by only 2-3% because most rows come from (18), (19), (23) and (37), which are kept.

4. Computational verification

Set-up. The corrected and reduced models are built from one row generator that also evaluates solutions, so the verified constraints are exactly the solved ones. Solver: COPT 8.0.7 (academic licence) on an Apple M5 (10 cores). Every COPT solution is decoded into routes and re-timed with the SVNS schedule evaluator. It is then checked for physical executability (launch and retrieval only at truck-visited nodes, sweep pairs flown consecutively, endurance) and against every row of the corrected model.

4.1 Check 1: the 105 SVNS plans satisfy all constraints

All 30 main-experiment plans and 75 fleet-size plans were mapped to MILP variable values and evaluated row by row. **No row of (2)-(43) is violated, in either the printed or the corrected model.** Figure 2 shows how often each constraint is active (its logical condition is switched on) and tight (it holds with equality). Removed constraints can be tight, e.g. (7), (8)-(11) and (26). Being tight does not make them necessary, since other constraints impose the same values. This is why the reduction relies on the proofs in Section 3 and the solver experiments below, not on tightness counts.

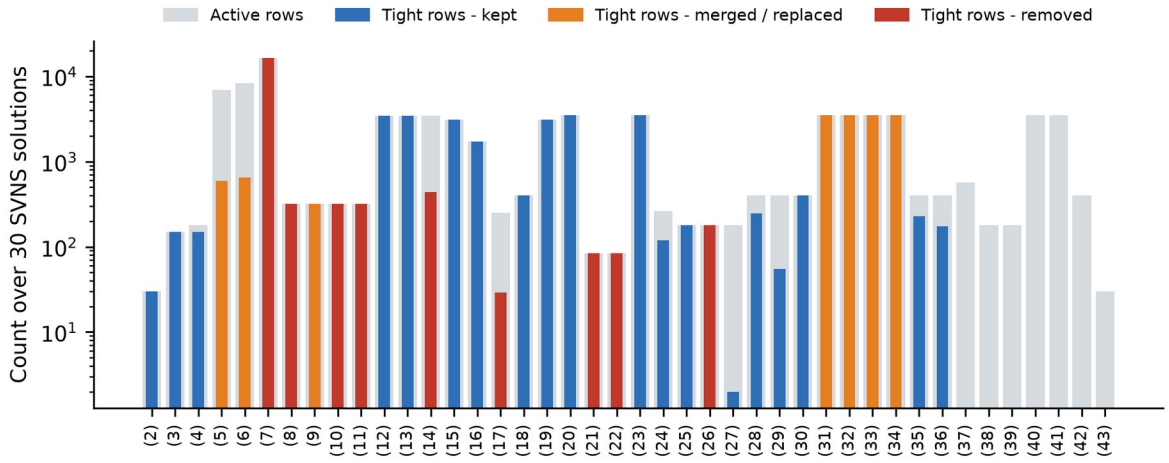


Figure 2. Active and tight rows of each constraint over the 30 SVNS plans (corrected model, log scale). Colours show the proposed treatment.

4.2 Check 2: optimality preserved on instances solved to proven optimality

Two small instances, TOY_1D (1 drone; area, line and point targets) and TOY_2D (2 drones), were solved to proven optimality (gap 0) under 16 configurations. Results are in Table 3 and Figure 3.

Table 3. Proven optimal objective values (✗ = decoded plan not executable).

Configuration	Rows 1D	Optimum TOY_1D	Rows 2D	Optimum TOY_2D	Conclusion
Printed model (no corrections)	1521	7.3782 ✗	609	7.0711 ✗	invalid (needs corrections)
Corrected model (reference)	1670	7.9084	662	7.2711	reference
Reduced model (proposal)	1530	7.9084	581	7.2711	**valid**
Corrected − (2)	1669	7.9084	661	7.2711	replaced by (2')
Corrected − (7)	1650	7.9084	650	7.2711	removable
Corrected − (8)-(11)	1646	7.9084	648	7.2711	removable
Corrected − (14)	1658	7.9084	654	7.2711	removable
Corrected − (17)	1597	7.9084	620	7.2711	removable
Corrected − (26)	1657	7.9084	655	7.2711	removable
Corrected − (21)(22), (12)(13) not extended	1668	7.4767 ✗	659	7.2711 ✗	only together with (12')(13')
Corrected − (16)	1667	7.8761 ✗	661	7.2711	must be kept
Corrected − (23)	1558	7.1711 ✗	635	7.0711 ✗	must be kept
Corrected − (37)	1366	7.8286 ✗	500	7.2711	must be kept

Corrected − (38)-(41)	1648	7.9084	644	7.2711	not binding here; keep (problem definition)
Corrected − (42)	1667	7.9084	658	7.2711	not binding here; keep (problem definition)
Corrected − (43)	1669	7.9084	661	7.2711	not binding here; keep (problem definition)

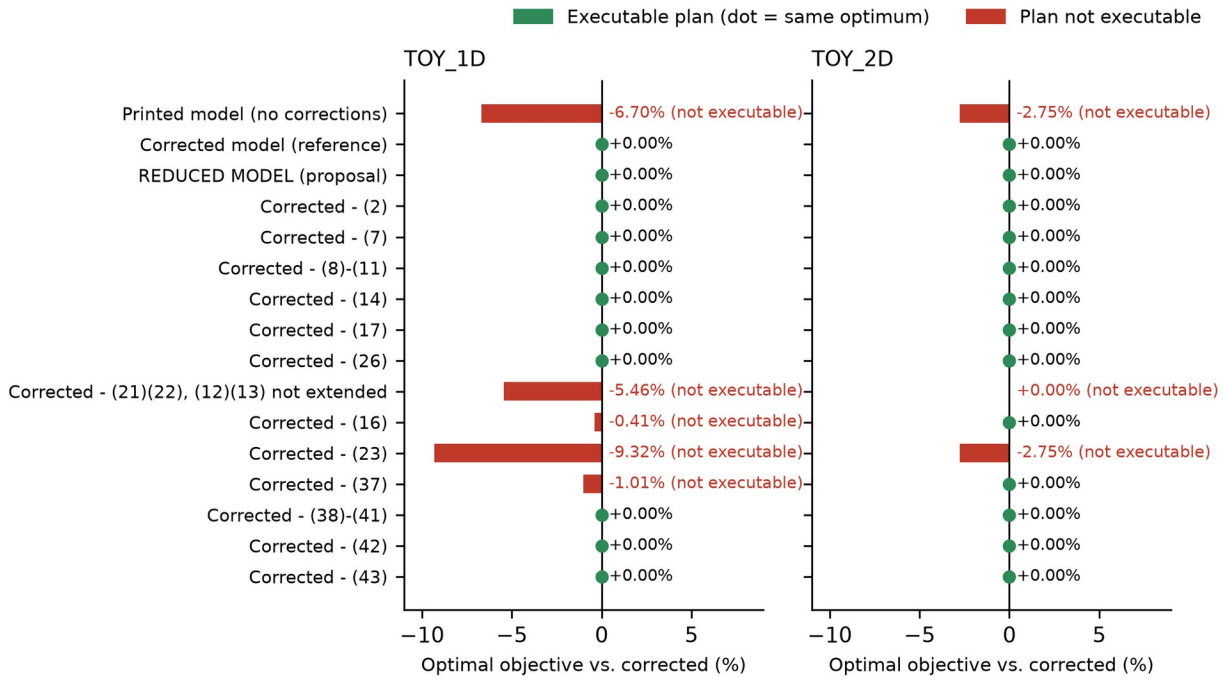


Figure 3. Deviation of the proven optimum from the corrected model. The reduced model and every removed constraint keep the optimum (green dots). Missing corrections, or removing (16), (23) or (37), give lower but unexecutable plans (red).

4.3 Check 3: the three small instances of the paper, 30-minute COPT runs

C-4-3, R-4-3 and RC-4-3 each have 6 operational nodes, 2 drones, one area target (8 coverage nodes in 4 sweep pairs), one line target and one point target. For each instance, the reduced and the full (corrected) models were solved in parallel for 1800 s (5 threads each), starting from the SVNS plan. For C-4-3 the 40-minute logs are evaluated at 1800 s.

Table 4. Results after 1800 s (same machine, same settings).

Instance	Model	Best objective	Lower bound	Gap	Gap ≤ 5% at	Gurobi (R1)
C-4-3	Full	7.4799	7.1843	3.95%	728 s	7.48
C-4-3	**Reduced**	7.4799	7.1840	3.96%	210 s	7.48
R-4-3	Full	7.4748	7.2320	3.25%	1030 s	7.47
R-4-3	**Reduced**	7.4748	7.2748	2.68%	656 s	7.47
RC-4-3	Full	7.9578	7.4043	6.96%	-	7.895
RC-4-3	**Reduced**	7.9140	7.3043	7.70%	-	7.895

Feasibility and optimal value. On C-4-3 and R-4-3 both models reach the Gurobi value reported in R1. On RC-4-3 the first 30-minute runs end at 7.9140 (reduced) and 7.9578 (full); a second 30-minute run of both models started from the 7.9140 plan ends at 7.9140 for both, 0.24% above the Gurobi value 7.895, while all lower bounds stay below 7.895. Every returned plan is executable when re-timed by the SVNS evaluator, and the full model accepts the reduced model's plan as a feasible start. The plans of the reduced model violate at most rows of removed constraints such as (7) or (14), and only in ways that do not affect routes or times (Sections 3.1 and 3.3). In an additional 300 s run on C-4-3 **without** a start solution, the reduced model found 7.4799 on its own.

Optimality. No run closed the gap within 30 minutes. The lower bounds of the two models are of the same size on every instance (equal on C-4-3, higher for the reduced model on R-4-3 and for the full model on RC-4-3), so the reduction does not systematically weaken the relaxation.

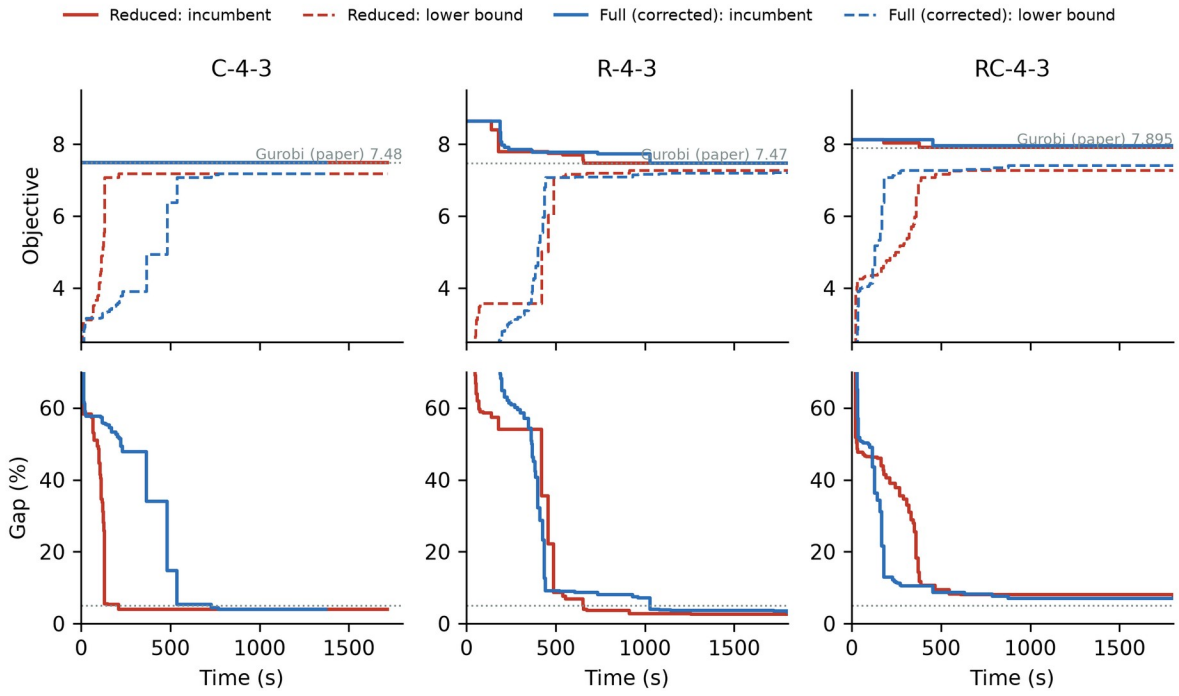


Figure 4. 30-minute runs on the three small instances: incumbent (solid), lower bound (dashed) and gap.

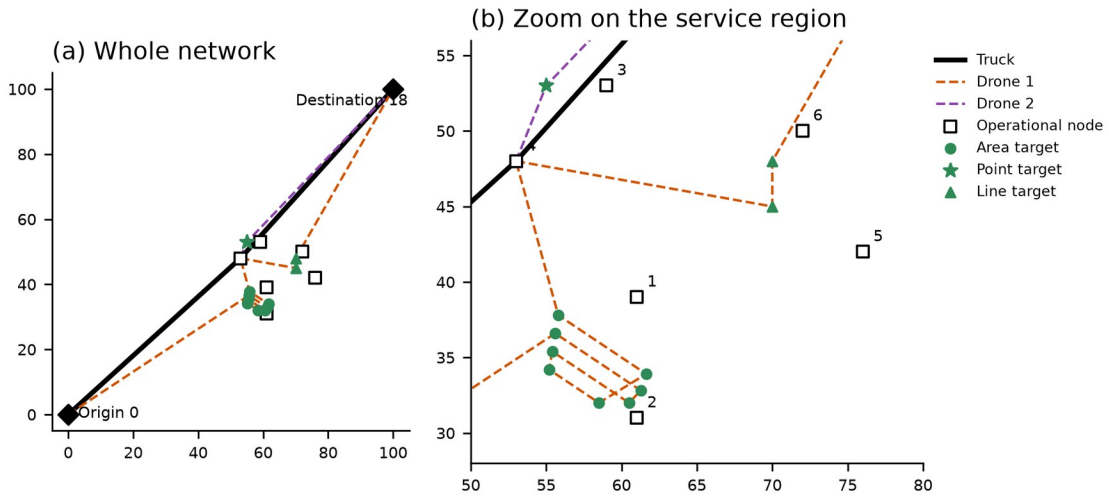


Figure 5. Plan returned by the reduced model for C-4-3 (identical to the corrected model's plan), objective 7.4799.

4.4 Efficiency: what can and cannot be claimed

Branch-and-bound running times vary from run to run. Table 5 and Figure 6 report all six COPT runs on C-4-3, including the shorter runs made earlier (longer runs are evaluated at 1800 s).

Table 5. All C-4-3 COPT runs (incumbent 7.4799 in every run). Run A's log stops at 652 s because of output buffering.

Run	Bound (gap) at 300 s	Gap ≤ 5% reached	Last logged gap
Reduced A: SVNS start, 10 thr	7.084 (5.3%)	351 s	3.77% at 652 s
Reduced B: SVNS start, 10 thr	4.947 (33.9%)	-	33.87% at 300 s
Reduced C: no start, 10 thr	7.208 (3.6%)	223 s	3.63% at 300 s
Reduced D: SVNS start, 5 thr	7.180 (4.0%)	210 s	3.96% at 1717 s
Corrected E: SVNS start, 10 thr	7.071 (5.5%)	-	5.47% at 300 s
Corrected F: SVNS start, 5 thr	3.904 (47.8%)	728 s	3.95% at 1369 s

In 3 of 4 runs, the reduced model reached a gap of 5% or less within 351 s. The full model did so only after 728 s, in 1 of 2 runs. The reduced model also processed about 2.2 times more nodes in the same time. This suggests faster bound improvement, but run B (reduced, still 33.9% after 300 s) shows that individual runs differ a lot. **We therefore present the reduction as a simplification that keeps the optimum and does not slow the solver, not as a proven speed-up.**

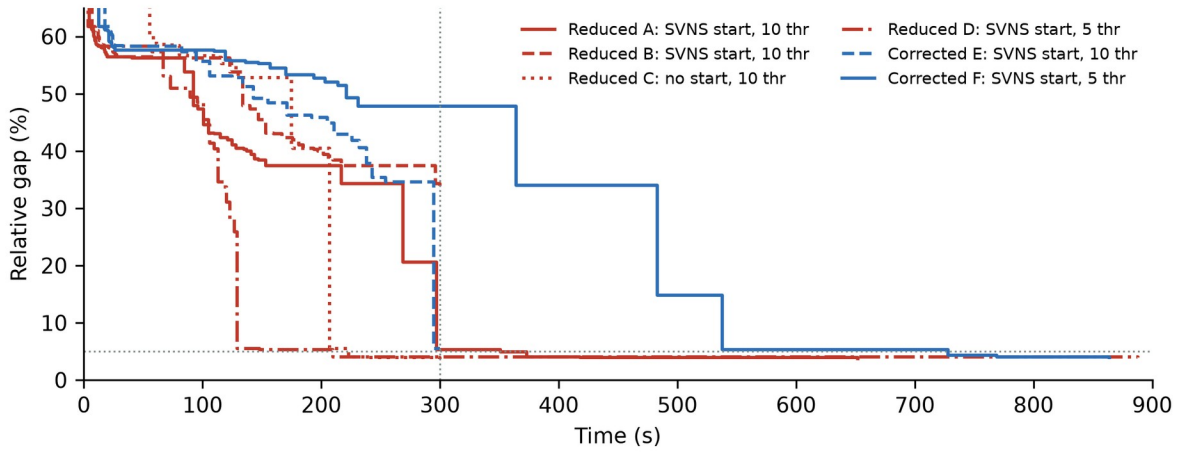


Figure 6. Gap over the first 900 s for all C-4-3 runs (red: reduced; blue: corrected/full).

4.5 Constraint activity in the C-4-3 COPT plan (corrected model)

Table 6 gives, for the constraints affected by the reduction, the active/tight counts in the C-4-3 COPT plan (re-timed to earliest times, same convention as for the SVNS plans). The counts agree with the SVNS statistics.

Table 6. Activity of the affected constraints.

Constraint	C-4-3 COPT plan (active / tight)	30 SVNS plans (active / tight)	Treatment
(7)	56 / 56	16,476 / 16,476	removed: implied by (5)-(6)
(8), (9), (10), (11)	2 / 2 each	317 / 317 each	(9) → (9'); others removed
(14)	10 / 4	3,432 / 440	removed: bounds not needed by MTZ
(17)	3 / 0	252 / 29	removed: never binding ($\lvert D \rvert \le \overline{v}$)
(21); (22)	1 / 1; 1 / 1	84 / 84; 84 / 84	removed: merged into (12')(13')
(26)	2 / 2	179 / 179	removed: waiting shifted to departure
(16)	5 / 5	1,716 / 1,716	kept (essential)
(23)	11 / 11	3,516 / 3,516	kept (essential)
(37)	1 / 0	571 / 0	kept (essential, though never tight)
(38); (40)	2 / 0; 11 / 0	179 / 0; 3,516 / 0	kept, merged; not binding here
(42); (43)	3 / 0; 1 / 0	401 / 0; 30 / 0	kept; not binding here

5. Conclusion

- Correct (2), (18) and (30) as in Section 2. These are genuine errors in the printed model; the original Gurobi implementation already contained the corrected forms.
- Replace the first-stage model by the 28-constraint formulation of Table 2. Remove δ_s^d and (47)-(49), and state $w_i^{sd} \in \mathbb{Z}_+$ in the variable list.
- Constraints (38)-(43) are not binding on the reported instances (time windows $[0, 50]$), but they are kept because they define the problem class.

Appendix A. Live progress dashboard (snapshots)

Snapshots of the live dashboard used to monitor the runs. The chart labels are in Chinese because the dashboard was built for internal monitoring.

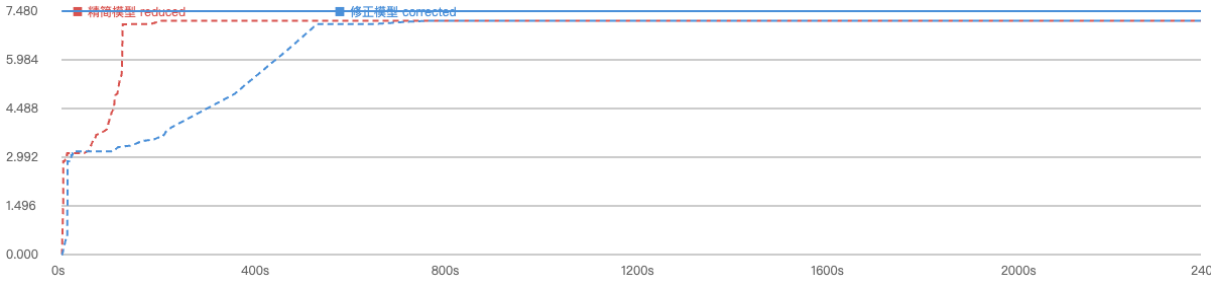


Figure 7. Dashboard, C-4-3: incumbent (solid) and lower bound (dashed).

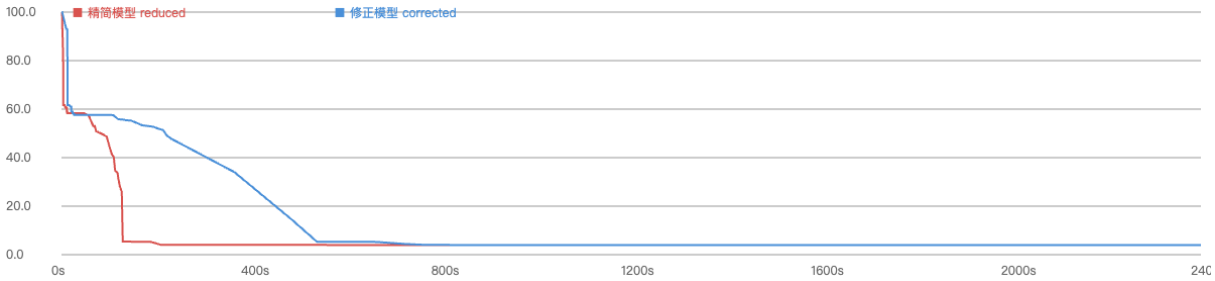


Figure 8. Dashboard, C-4-3: gap (%).

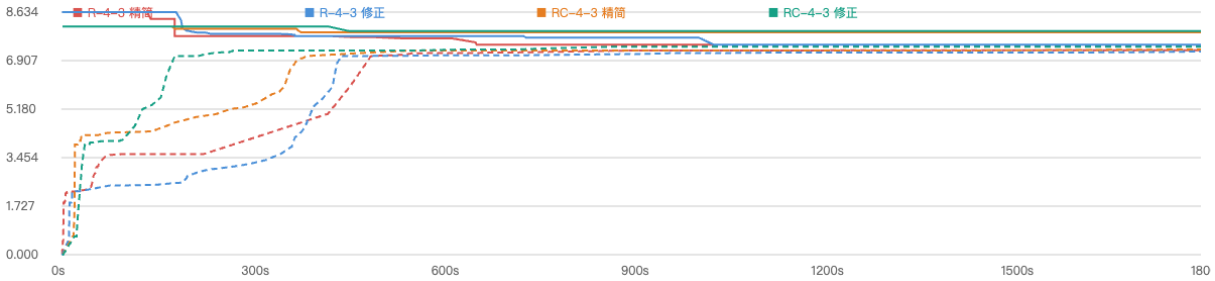


Figure 9. Dashboard, R-4-3 and RC-4-3: incumbent and lower bound.

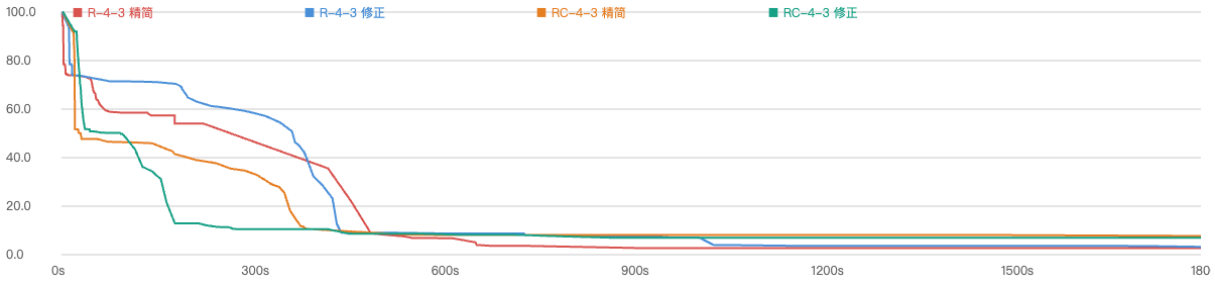


Figure 10. Dashboard, R-4-3 and RC-4-3: gap (%).

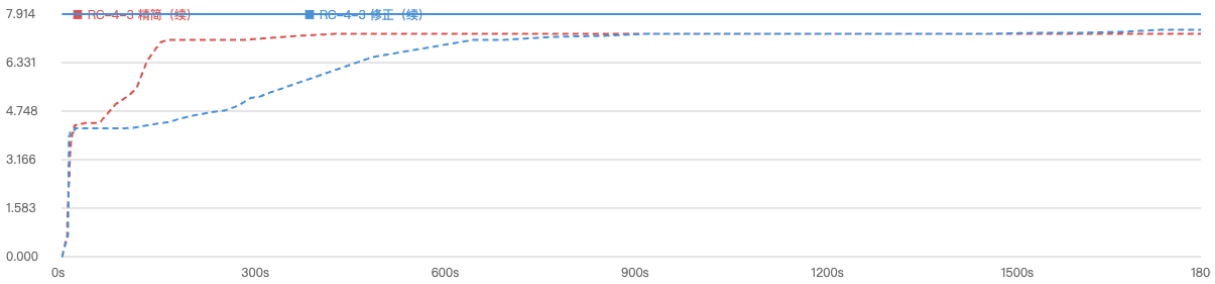


Figure 11. Dashboard, RC-4-3 second run from the 7.9140 plan: incumbent and lower bound.

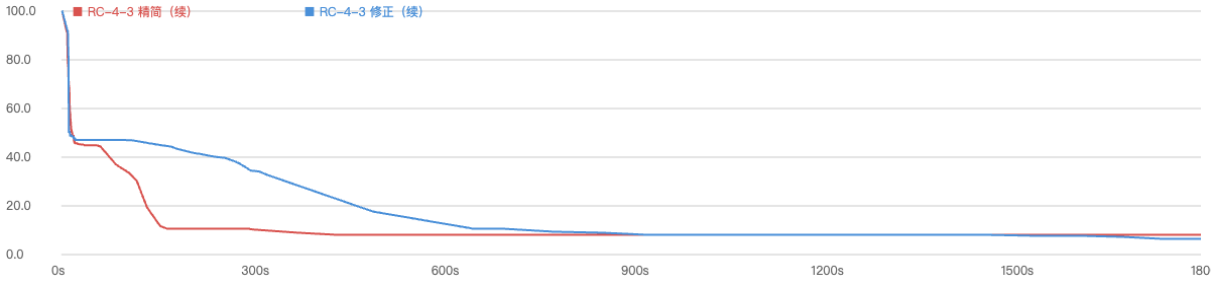


Figure 12. Dashboard, RC-4-3 second run: gap (%).

Appendix B. Reproducibility

All scripts and data are in the folder constraint_audit/ of the repository: model_rows.py (row generator for (2)-(43) and the three model variants), copt_model.py and run_copt_small.py (COPT runs with post-solve checks), run_copt_experiments.py (toy instances), run_svns_audit.py (SVNS audit), progress_dashboard.py (live dashboard), make_report_figures.py, make_report_docx.py and omml.py (this report). Logs are in copt_logs/ and outputs/, full solutions in copt_runs/.